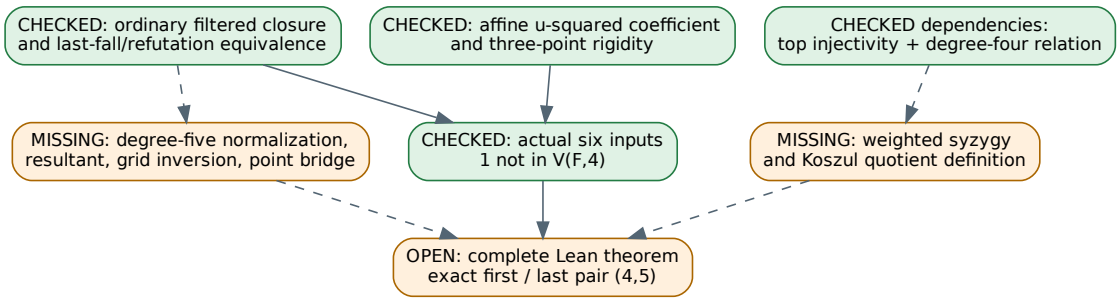
Guarded fall degrees: Lean proof progress

2026-10-29 — Exact-pair target remains open

The requested theorem is the exact ordinary first/last fall pair $(4, 5)$ for every unsatisfiable inverse-guarded three-abscissa short-Weierstrass system under the study's rational-point and characteristic hypotheses.

New verified theorem. For every field K with $6 \neq 0$, every three distinct $a, b, c \in K$, and arbitrary curve coefficients and target abscissa, `GuardedModel.no_degree_four_refutation` proves $1 \notin V(F, 4)$ for the actual six generators. It uses neither recorded ranks nor a hypothesis asserting the lower bound. With an explicit unit-ideal hypothesis, `lastFall_lower` excludes all last-fall bounds below five. The field-of-definition/unit-ideal bridge is still open.

Structural proof. Ordinary closure is defined on Mathlib multivariate polynomials by finite derivations, including arbitrary multipliers when the actual product has degree within the bound. Its least-subspace property and the last-fall/refutation equivalence for unit ideals are checked. The separating space is $I + Kf_1 + Kf_2$, where I vanishes on the guarded base grid while u remains formal. Its degree-at-most-three part lies in I: the affine u^2 coefficient cannot interpolate nonzero quartic weights on three distinct points. Degree additivity proves full multiplicative closure at four. The space excludes one by grid evaluation.



Additional checked dependencies. The quadratic divided difference is actually derived inside $V(F, 4)$ with polynomial degree bounds. Low-degree top-map injectivity, the degree-four homogeneous relation and its lift are checked. A weighted syzygy/Koszul formalization is still required before claiming the numeric first-fall equality.

Reproduce. Pinned Lean 4.34.1, Mathlib `d13f23b723b8a846827a245b89c10fc7d3f11612`. Run `lake build`, the repository's `AuditAxioms.lean` on all six new modules, and fresh `leanchecker` on `FirstFallCore`. Exact commands, logs, hashes and hypotheses are in `exact-pair-progress.md` and `exact-pair-validation/`. Only standard Lean foundations are accepted; no custom theorem axioms, compiler-trusting evaluation or placeholders are used.

Remaining scope. The universal degree-five normalization, resultant, finite-grid inversion and rational-point bridge are not yet formalized. Thus the requested exact pair is partial. No new experiments, timings, growing-base or characteristic-two degree claims are made. Sources: Huang–Kosters–Yang–Yeo, arXiv:1505.02532, Definitions 2.1–2.2; the study's symbolic and independent audits; pinned Mathlib sources.